

High-Performing Deep Learning Regression Models for Predicting Low-Pressure CO₂ Adsorption Properties of Metal–Organic Frameworks

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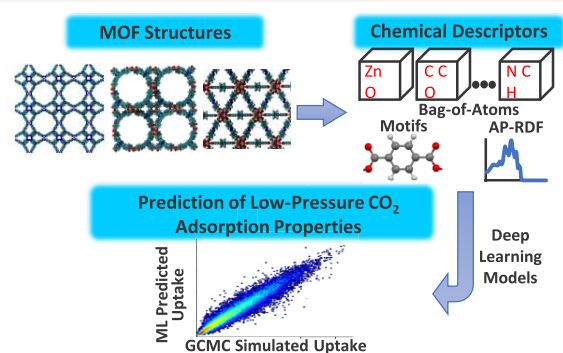


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ABSTRACT: Metal–organic frameworks (MOFs) have garnered interest as potential solid sorbent materials for postcombustion CO₂ capture. With a seemingly infinite design space, high-throughput computational screening of MOFs has developed into an effective tool for the development of new materials. In this work, machine learning (ML) has been used to develop accurate quantitative structure–property relationship (QSPR) models to rapidly predict the CO₂ working capacity and CO₂/N₂ selectivity at the low-pressure conditions relevant to postcombustion carbon capture (0.15 bar CO₂, 0.85 bar N₂). A database of over 340 000 MOFs constructed from hundreds of types of building units arranged in over 1000 net topologies was used to train and test the models. Neural network ML models were optimized using six geometric descriptors along with three so-called chemical descriptors, namely, the atomic property-weighted radial distribution function (AP-RDF) and some variants thereof, the bag-of-atoms, and the chemical motif density descriptors. The ML models built using geometric descriptors alone resulted in test set correlation R^2 values of only 0.71 and 0.75 for CO₂ working capacity and CO₂/N₂ selectivity, respectively. ML models built with a single type of chemical descriptor all outperformed the geometry-only models giving R^2 values ranging from 0.83 to 0.94 with the AP-RDF model being the most accurate. Overall, the best model was built using a combination of AP-RDF, chemical motif, and geometric descriptors (R^2 = 0.96 when predicting the CO₂ working capacity and R^2 = 0.95 for the selectivity). To date, these are the most accurate ML models for predicting low-pressure gas uptake of MOFs. The combined model was able to capture 994 of the true top 1000 MOFs (from a test set of ~70 000) within the top 5000 MOFs as predicted by the model with CO₂ working capacity as the target. Thus, if the ML model were used to prescreen materials for more compute intensive GCMC simulations, then it would result in a greater than 10 times speed up while still capturing >99% of high-performing materials. These results highlight the importance of chemical descriptors in predicting low-pressure gas adsorption properties in nanoporous materials.



INTRODUCTION

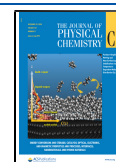
Metal–organic frameworks (MOFs) are a diverse class of materials built from organic and inorganic building units that can self-assemble to form nanoporous frameworks. Since they can be constructed from a seemingly infinite number of combinations of building units, the hallmark feature of MOFs is the potential to tune the materials for a given application. To date, MOFs with an expansive range of pore sizes, geometries, surface areas, and chemistries have been synthesized for a wide range of applications.^{1–5} One area in which MOFs have attracted significant attention is as solid sorbents for gas separations and storage applications.^{6–9} For example, MOFs have been proposed to be used as a medium for vehicular natural gas storage where methane is adsorbed at high pressures (~35 bar).^{10–12} MOFs have also received attention as sorbent materials for postcombustion CO₂ capture and storage (CCS),^{13–17} where it is proposed they could be used in pressure swing adsorption (PSA) systems retrofitted to power

plants burning fossil fuels. In such a system, the CO₂ is selectively adsorbed from the flue gas by a packed bed of the MOF. Once the MOF bed is saturated, a vacuum is applied to desorb the captured CO₂ and regenerate the sorbent. In addition to or instead of applying a vacuum, the temperature of the MOF bed may be raised to promote regeneration of the sorbent. The desorption process ideally yields >95% pure CO₂ that can then be compressed for transport and permanent storage. It is generally believed that the preferable gas adsorption properties of the MOF for postcombustion CCS are high CO₂/N₂ selectivity and high CO₂ working capacity,

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where the working capacity is the difference between the CO₂ uptake at the adsorption conditions compared to at the desorption conditions.^{18–20} It is notable that for postcombustion CCS the CO₂ must be selectively adsorbed at a relatively low partial pressure of ~0.15 bar as compared to many other gas separation and storage applications where the adsorption typically occurs at much higher partial pressures.

Atomistic modeling methods, in particular grand canonical Monte Carlo (GCMC) simulations, can provide generally accurate adsorption properties of a MOF if the structure is known.²¹ As such, high-throughput screening techniques have been developed to screen large libraries of both lab-synthesized materials and computer-generated hypothetical materials.^{22,23} Wilmer and Snurr were the first to do this when they screened a library of ~140 000 hypothetical MOFs for methane storage at 35 bar.²⁴ The simulations revealed that MOFs with pore sizes of 4 and 8 Å, which accommodated one or two methane molecules, respectively, resulted in high storage capacities. Since then, many large databases of materials have been screened,^{25–28} leading to the generation of substantial amounts of data that can be used to develop quantitative structure–property relationship (QSPR) models of gas adsorption in MOFs with data-driven machine learning (ML) methods. ML models have the advantage that once trained the models can screen materials in a matter of seconds per MOF versus hours in the case of molecular simulation. Thus, accurate ML models could be used as part of a prescreening protocol to filter out poorly performing materials and thereby reduce the number of computer-intensive GCMC simulations that are performed during high-throughput screening.^{29–31}

In a very basic sense, ML algorithms work by generalizing patterns in example data and making predictions from these generalizations. Deep learning models, a subcategory of ML algorithms, are those that utilize artificial neural networks and are now being widely deployed in many areas of science. In the context of MOFs, ML models can be used to learn how to relate chemical or structural features of materials to their performance. We will refer to these features as *descriptors* for the ML models. Geometric descriptors, such as surface area, pore size, and density have been used extensively in the literature to predict gas uptake capacity in nanoporous materials using machine learning.^{30,32–34} While these geometric descriptors correlate well with adsorption properties at higher pressures, they perform considerably worse at lower pressures.^{31,35} One explanation for this observation is that at high pressure the adsorption is essentially a packing problem, so the geometry of the MOF can be correlated well to the adsorption properties. However, at low pressures, the chemical interactions of the gas with the MOF framework are likely to be more important in determining the adsorption properties. Therefore, “chemical” descriptors, or those that somehow capture the chemistry of the MOF pores, are anticipated to be more relevant in ML models for predicting low-pressure gas adsorption.

Perhaps the most straightforward chemical descriptor is the chemical motif density of a MOF. For example, if a guest molecule is known to bind to a given functional group, then the functional group count per unit volume in the MOF should be a relevant descriptor. Another related chemical descriptor is the so-called “bag-of-atoms” descriptor. The bag-of-atoms descriptor is an atom-type volume density weighted by the van der Waals radius or the interaction strength of each atom-type or element. The interaction strength of an atom could be taken

from the well-depth of the Lennard-Jones parameters of the atom type. Anderson et al. found that the chemical motif density and bag-of-atoms descriptors improved prediction of adsorption isotherms of MOFs when used in conjunction with standard geometric descriptors when studying the adsorption of various alchemical adsorbates.³⁴

Another descriptor that captures chemical information (as well as geometric) is the atomic-property-weighted radial distribution function (AP-RDF) developed by Woo and co-workers.³²

$$\text{RDF}^P(R) = f \sum_{i,j}^{\text{atom pairs}} P_i P_j e^{-B(r_{ij}-R)^2} \quad (1)$$

The form of the AP-RDF is given in eq 1, where r_{ij} is the minimum image convention distance of all atom pairs in the MOF, P_i is an atomic property of the atom i , B is a Gaussian smoothing parameter, and f is a normalization factor. The atomic property, P , imbues the descriptor with chemical information and can be any the tabulated property of the atoms, such as the van der Waals radius or electronegativity. The descriptor can be viewed as a “distance-between-features” descriptor where the assumption is that a guest molecule’s interaction with a pair of framework atoms in the MOF will depend on the atom pair’s distance and its chemical properties. This is depicted schematically in Figure 1. For each property,

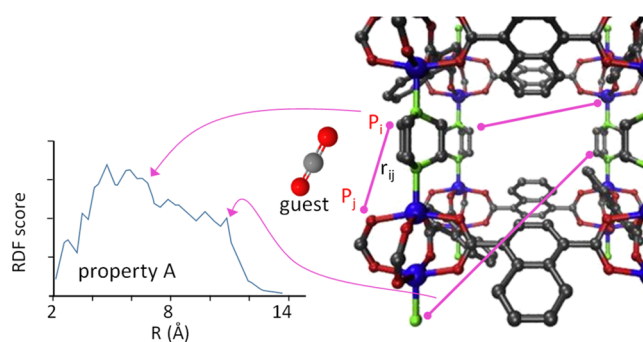


Figure 1. Schematic representation of how the AP-RDF descriptor is constructed to form a fingerprint for a MOF. The value of the AP-RDF at a specified distance for each property is used as a single descriptor.

the AP-RDF gives probability of finding atom pairs inside the unit cell of the MOF of a given distance that is weighted by the atomic properties. Therefore, the AP-RDF provides a unique fingerprint for each MOF that can be used for machine learning. Variations of the AP-RDF can also be envisioned. For example, one could replace the product of each atomic property in eq 1, with the difference in the properties, or one could replace the summation in the equation with a maximum function. Although these variations of the AP-RDF descriptor have the potential to encode different information compared to the original, to date, they have not been explored. Several advantages of the AP-RDF descriptor over other chemical descriptors reported in the literature are as follows: (1) The AP-RDF carries both geometric and chemical information. (2) The AP-RDF is not specific to a database on which it is trained (cf. chemical motifs). (3) The AP-RDF is invariant to rotations, translations, and replications of the MOF unit cell (cf. bag-of-atoms).

The first application of the AP-RDF was to categorize the CO₂ uptake capacity of MOFs at low pressure (0.15 bar). Using the support vector machine algorithm, Fernandez et al. built ML classifiers that could be used to sort MOFs into two categories: high- and low-performing.³⁶ The model was trained and tested on a hypothetical database of ~320 000 MOFs for which GCMC simulations had been run to determine their CO₂ uptake capacities. It was found that the classifier could recover 94% of the top-performing materials while flagging only 10% of the total database as high-performing. These MOFs would then be selected for more computer-intensive screening with atomistic simulations. Interestingly, using the same database of materials, Fernandez utilized more advanced ML algorithms to build accurate classifiers for low-pressure CO₂ uptake based only on geometric descriptors.³⁷ More recently, Dureckova et al. used the AP-RDF and geometric descriptors to build QSPR models for predicting CO₂ adsorption properties relevant to high-pressure (40 bar) precombustion CO₂ capture.³⁰ In that work, regression models were built to provide a prediction for the CO₂ working capacity and CO₂/H₂ selectivity, rather than just classifying the MOFs as high- or low-performing. It was found that ML models built using the AP-RDF descriptors were outperformed by those built from only geometric descriptors, with correlation $R^2 = 0.86$ for the AP-RDF and $R^2 = 0.89$ for the geometric models. This is perhaps not surprising since the adsorption is at a very high pressure where the geometric factors are expected to dominate. It is notable, however, that models built using both kinds of descriptors provided significantly improved results with $R^2 = 0.94$.

Another type of descriptor that has emerged to predict adsorption properties in MOFs is not a purely structure-based descriptor in that it involves running a short atomistic simulation using a simple probe “atom” to derive the descriptor.^{35,38} Fanourgakis et al. developed one such descriptor for predicting gas adsorption capacities of nanoporous materials for which there are non-negligible electrostatic interactions between the gas and the framework (e.g., CO₂).³⁵ They used probe atoms (called Dprobes) of varying van der Waals radii and centered electric dipoles for which a short GCMC simulation is run to construct the descriptor. While the simulation used to derive the Dprobes descriptor is short compared to a full GCMC simulation, the simulations require partial atomic charges to be derived for each MOF. However, deriving quality partial atomic charges can often be the bottleneck in a GCMC workflow. Using this descriptor, they performed machine learning on a subset of the CoRE MOF database,³⁹ containing 2932 structures. At a pressure of 0.1 bar, they were able to predict CO₂ uptake with an R^2 value of 0.83 when the Dprobe descriptor was combined with geometric descriptors.

Since the early work of Fernandes et al.³⁶ of using the AP-RDF descriptor to predict low-pressure CO₂ adsorption in MOFs, several developments have occurred that make it worthwhile to revisit the problem. First, and most importantly, sophisticated algorithms to generate hypothetical materials have been developed such that significantly more diverse databases of nanoporous materials now exist.^{23,40,41} The database first used by Fernandez³⁶ contained ~324 000 hypothetical MOFs that were constructed using 7 inorganic structural building units (SBUs), 59 organic SBUs, and 19 functional groups. That database spanned less than 20 different network topologies with most structures having the simple

cubic topology. In contrast, the newer database used by Dureckova et al.³⁰ was constructed from 23 inorganic SBUs, 175 organic SBUs, and 50 functional groups and contains MOFs with over 1166 network topologies. Additionally, significantly more advanced ML algorithms, including deep learning methods, have become more accessible to researchers since that initial work. Furthermore, accurate regression models built from purely structural descriptors still have not been developed for low-pressure CO₂ uptake, and it remains an open question as to whether chemical descriptors are needed to build accurate models of gas uptake at low pressure.

In this work we explore whether accurate ML regression models can be built to predict the low-pressure CO₂ adsorption properties using a combination of geometric descriptors and chemical descriptors, namely, the bag-of-atoms, chemical motif density, and the AP-RDF. With the hypothesis that the chemistry of the MOF pores is important for low-pressure adsorption of CO₂, we intend to explore the relevance of chemical and geometric descriptors to this problem. We also intend to explore the relevance of different variations of the AP-RDF descriptor. All of the models will be trained and tested on the large and diverse database of over 340 000 nanoporous materials first introduced by Dureckova et al.

This paper is organized as follows: We first describe the MOF database, followed by the computational methods and procedures. The third section of the manuscript will summarize the results and discussion, which will be followed by a summary of the conclusions.

METHODS

MOF Database. Calculations were carried out on a diverse set of 342 489 hypothetical MOF structures. The MOFs are constructed from a total of 23 inorganic and 175 organic SBUs, arranged in 1166 net topologies and functionalized with 50 different groups. MOF structures were generated using the ToBasCCo algorithm which uses underlying net topologies and a list of SBUs to generate a larger, more structurally diverse database of MOFs.⁴⁰ Geometries of all MOFs were optimized using the Universal Force Field (UFF),⁴² where the positions of the atoms covalently bound to metal atoms were frozen. The database is provided as part of the [Supporting Information](#).

GCMC Simulations. The CO₂ working capacity and CO₂/N₂ selectivity that were target properties for machine learning were determined from GCMC simulations. The adsorption properties of CO₂ were determined from a binary mixture of CO₂/N₂ at partial pressures of 0.15 and 0.85 bar, respectively, at 298 K. For regeneration, a temperature/pressure swing (TPSA) system was assumed with a desorption temperature of 298 K and vacuum pressure of 0.03 bar assuming a 99% CO₂, 1% N₂ mixture. A minimum of a 3 × 3 × 3 supercell of each MOF was used for the GCMC simulations which were run for 40 000 cycles, where a cycle corresponds to one Monte Carlo step for each guest molecule in the simulation cell. We used 20 000 cycles for equilibration and 20 000 cycles for production. Interactions between the guest and MOF were modeled using atom-centered Lennard-Jones (LJ) potentials and point-charge Coulomb potentials. A 12.5 Å cutoff was used for the LJ potentials, while the Ewald summation method was used to treat the long-range Coulomb potentials. For CO₂, the nonbonded parameters and partial atomic charges were taken from Garcia-Sánchez et al.,⁴³ while for N₂ the parameters

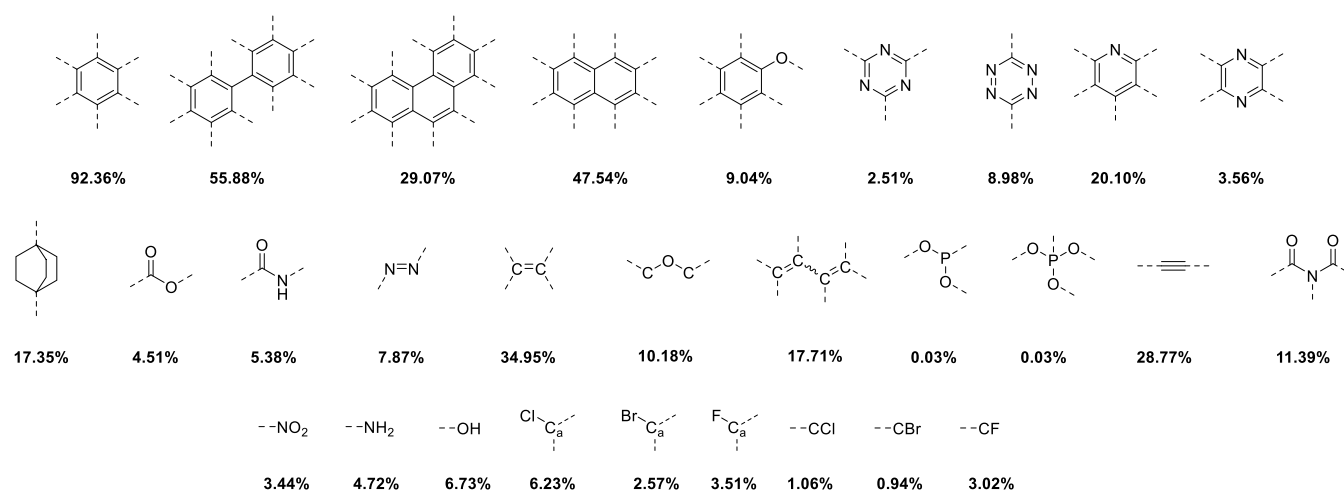


Figure 2. Chemical motifs used in the calculation of the chemical motif descriptor. The percentages represent the frequency of each motif in the database. C represents an aliphatic carbon and C_a represents an aromatic carbon.

were those from Provost⁴⁴ that were fit to reproduce N₂ isotherms in MOFs. For the MOF framework atoms, LJ parameters were taken from the UFF force field^{42,45} and Lorentz–Berthelot mixing rules were used to combine the LJ parameters. Partial atomic charges on the MOF framework atoms were assigned using the MEPO-SQE method⁴⁶ which is an empirical model that was parametrized to reproduce the DFT-derived electrostatic potential charges in MOFs using the REPEAT method.⁴⁷

Descriptor Evaluation. The Zeo++ program was used to compute the geometric descriptors, namely, the volumetric and gravimetric surface areas, void fraction, density, dominant pore size, and largest accessible pore size.⁴⁸ The AP-RDF descriptor as given by eq 1 was determined using the smoothing parameter $B = 10$ as recommended by Fernandez et al.³² Other values were tested but the results were found to either give similar or lower performing results. The atomic properties used in this work were the elements' electronegativity, hardness, and van der Waals volume whose values are provided in the [Supporting Information](#). The function was normalized by the number of framework atoms in the cell and scaled by a factor of 0.001 (i.e., $f = 0.001/N_{\text{atoms}}$). The AP-RDF descriptors were calculated using a distance range of 2.0–30.0 Å with 113 steps that linearly increase from 0.004425–0.5 Å. This gave 339 AP-RDF descriptor values for each MOF. The variants of the AP-RDF descriptors defined in this work were calculated using a distance range of 2.0–30.0 Å in 113 steps with a constant step size of 0.25 Å. The variants of the RDF descriptor were calculated according to eq 1, but in one case the difference between properties was taken instead of the absolute value of the product between properties, termed the RDF (difference) descriptor. In another case, the summation was replaced with a maximum function, termed the RDF (maximum) descriptor. The descriptor calculated according to eq 1 is termed the AP-RDF, or RDF (product) descriptor. The combination of these three descriptors is termed the RDF (combination) descriptor.

The chemical motifs descriptor was calculated as described by Anderson et al.,³⁴ however, this descriptor has been used frequently in the literature by other researchers.^{31,33,34} The motifs used in the calculation of this descriptor are shown in [Figure 2](#) along with the percentage of MOFs that contain the structural motif. The bag-of-atoms descriptor was calculated according to the procedure outlined by Anderson et al.³⁴

Specifically, the unit cell of each MOF was split into $6 \times 6 \times 6$ cuboids, and the sum of the framework atom ϵ and σ parameters as defined in the UFF force field was computed for each cuboid. Each descriptor value was then normalized by the total number of framework atoms in the MOF. As such, each MOF was split into 216 cuboids with two parameters per cuboid giving a total of 432 descriptor values per MOF.

Neural Network Models. Artificial neural networks (ANNs) were created using the PyTorch deep learning framework due to its syntactic simplicity, capacity for fast experimentation, and versatility.^{49,50} The ANNs used in this work are fully connected feedforward neural networks, or more specifically multilayer perceptrons (MLPs), with a single output node in each to predict as the target value either CO₂ working capacity or CO₂/N₂ selectivity. The training process for all models involved first optimizing hyperparameters of the model where the total set of 342 489 MOFs was randomly split into training, development, and test sets in a 70:10:20 ratio. This process was carried out by training on a training set while simultaneously evaluating the hyperparameter set on a development set, where the mean squared error was used as the cost function. The optimization of the cost function was carefully monitored to determine the optimal number of epochs such that overfitting to the training data was avoided. Specifically, the value of the cost function of the development set as a function of the number of epochs was visually monitored to determine when the cost function plateaued and therefore determine the number of epochs required to complete training. An epoch is defined as the number of complete passes of the training data set through the network. The hyperparameters for each neural network model were optimized individually using an automated grid search. Once the hyperparameters were optimized, the final model was evaluated on the test set. More information on the optimization of the neural network models as well as the hyperparameter values for each model are provided in the [supporting material](#).

RESULTS AND DISCUSSION

Neural network models were built to predict the GCMC-computed CO₂ working capacity (WC) and CO₂/N₂ selectivity using 6 geometric descriptors (Geo), the bag-of-atoms (BOA), chemical motif density (CMD) and AP-RDF

Table 1. Performances of the Deep Learning Models for CO₂ Working Capacity and CO₂/N₂ Selectivity Built from Variants of the AP-RDF Descriptor^a

target	descriptors	Pearson R^2			Spearman R^2	RMSE ^b
		training	development	test		
CO ₂ working capacity	RDF (maximum)	0.947	0.848	0.845	0.866	0.25
	RDF (difference)	0.973	0.920	0.918	0.926	0.18
	RDF (product)	0.965	0.934	0.932	0.939	0.16
	RDF (combination)	0.966	0.939	0.936	0.940	0.16
CO ₂ /N ₂ selectivity	RDF (maximum)	0.936	0.841	0.839	0.889	16
	RDF (difference)	0.963	0.911	0.906	0.940	13
	RDF (product)	0.959	0.926	0.924	0.953	11
	RDF (combination)	0.993	0.940	0.937	0.960	10

^aProduct, difference, and maximum, as well as their combination, are as defined in the “Methods” section. The Pearson R^2 values on the training, development, and test datasets are given, while the Spearman rank R^2 and root mean square error (RMSE) on the test data set are given. AP-RDFs in this table were calculated with a static step size of 0.25 Å from 2 to 30 Å. ^bThe CO₂ working capacity RMSE values are in mmol/g, while the selectivity is unitless.

Table 2. Performances of the Deep Learning Models for CO₂ Working Capacity and CO₂/N₂ Selectivity Built from Various Combinations of Geometric (Geo), AP-RDF (RDF), Bag-of-Atoms (BOA), and Chemical Motifs (CMD) Descriptors^a

target	descriptors	Pearson R^2			Spearman R^2	RMSE ^b
		training	development	test		
CO ₂ working capacity	Geo	0.717	0.705	0.706	0.792	0.35
	CMD	0.879	0.838	0.838	0.870	0.26
	BOA	0.976	0.867	0.864	0.883	0.24
	RDF	0.974	0.938	0.936	0.944	0.17
	Geo+RDF	0.979	0.955	0.953	0.960	0.14
	Geo+CMD+RDF	0.993	0.960	0.958	0.965	0.13
CO ₂ /N ₂ selectivity	Geo	0.758	0.754	0.749	0.886	20
	CMD	0.888	0.832	0.829	0.903	17
	BOA	0.970	0.846	0.845	0.881	16
	RDF	0.962	0.928	0.926	0.956	11
	Geo+RDF	0.973	0.945	0.943	0.971	10
	Geo+CMD+RDF	0.982	0.952	0.948	0.975	10

^aThe Pearson R^2 values on the training, development, and test datasets are given, while the Spearman rank R^2 and root mean square error (RMSE) on the test dataset are given. ^bThe CO₂ working capacity RMSE values are in mmol/g, while the selectivity is unitless.

descriptors, and combinations of the descriptor types. Models were trained on a training set of 239 742 MOFs (70% of the database), and the hyperparameters evaluated on a development set of 34 249 MOFs (10% of the database). The optimized models were then evaluated on a test set of 68 497 MOFs (20% of the database) that were never used to train or develop the models. The performance of the models is summarized in Tables 1 and 2 which shows the Pearson correlation coefficient (R^2), the Spearman rank correlation coefficient (Spearman rank R^2), and the root-mean-square error (RMSE) of the models. R^2 measures the linear correlation between the actual and predicted targets. The model will inevitably be fit more closely to the training data than the development and test data because this is the data used to optimize the weights and biases of the model. However, the model performance between the development or test data set should be nearly the same assuming a reasonable split of the total data set. The Spearman rank correlation R^2 represents the linear correlation between the actual and predicted targets after the target values have been numerically ranked. Finally, the root-mean-square error (RMSE) represents on average how far the predicted values are from the actual values in the given units of the target property. The models with the highest R^2 on the test set and the lowest RMSE are deemed the best performing. We note that other ML

algorithms, specifically support vector regression and gradient boosted decision trees models, were tested on the same data set but were outperformed by the neural network models presented here in all cases.

A first objective of this study is to evaluate variations of the AP-RDF descriptor and identify the best form of the descriptor for predicting low-pressure CO₂ adsorption properties. The results of this investigation is summarized in Table 1. The original AP-RDF is defined in eq 1 and will be referred to as the RDF (product) descriptor in Table 1. The variant of the AP-RDF in which the product between properties is replaced by the absolute value of the difference between properties (i.e., $|P_i - P_j|$) will be referred to as the RDF (difference) descriptor. Finally, the variant of the AP-RDF in which the summation is replaced by a maximum function will be referred to as the RDF (maximum) descriptor. Additionally, a combination of all three descriptors was investigated to determine whether each descriptor carries complementary information. It should be noted that B values of 5, 10, 15, and 50 Å⁻² (eq 1) were evaluated for each variant, but it was found that as long as the B value was not less than 10 Å⁻² (too much smoothing) the results were unaffected. It is also important to note that while an upper limit for B was not found very large values of B would logically be unfavorable as well. Thus, the previously recommended value of 10 Å⁻² was used for all variants.

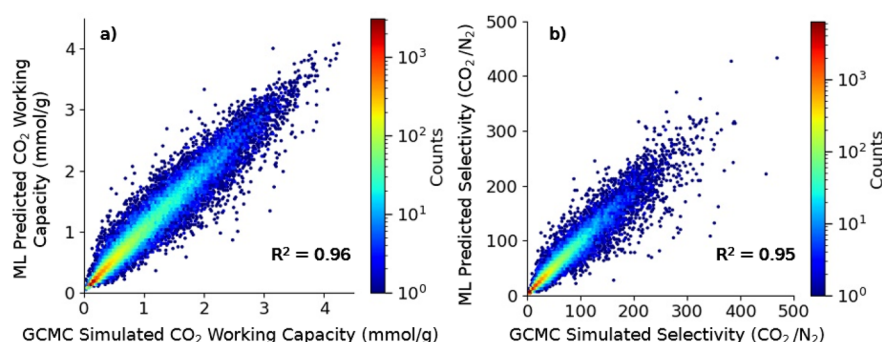


Figure 3. Heat maps of the best neural network model's predicted (a) CO_2 working capacity or (b) CO_2/N_2 selectivity as a function of the GCMC simulated property for the test set data. The model was built using geometric, AP-RDF, and chemical motifs descriptors. The color of the heat map corresponds to the density of data points with highest density to lowest density being represented by the range of red to blue, respectively.

Table 1 shows that the original product AP-RDF descriptor performs the best with test set $R^2 = 0.932$ for the WC and test set $R^2 = 0.924$ for the selectivity. The difference variant performs slightly worse with $R^2 = 0.918$ (WC), and $R^2 = 0.906$ (selectivity), while the maximum RDF performs significantly worse with $R^2 = 0.845$ (WC) and $R^2 = 0.839$ (selectivity). Although the variants do not perform better than the original product RDF, they may encode different information. To test this, models built with the combination of all three RDF descriptors were evaluated. However, the combination models showed a very modest improvement over the original product RDF alone with $R^2 = 0.936$ (WC) and $R^2 = 0.937$ (selectivity), suggesting that the RDF variants encode very little additional information. Since the combination model uses three times the number of descriptors (1017 vs 339) as the original RDF, when considering the balance between complexity and performance we chose to only continue with the original product RDF for the remainder of this work. We did, however, further optimize the product RDF by examining using smaller distance bins at closer distances, and larger distance bins at long distances. The hypothesis here is that the guest host interactions will be more sensitive to changes at short atom pair distances (i.e., 2 Å), compared to long distances of say 20 Å. The product AP-RDF descriptors calculated using a distance range of 2.0–30 Å in 113 steps that linearly increased from 0.004425–0.5 Å were compared to those where the 113 steps were held constant at 0.25 Å. Only a very modest improvement was found with the variable step size giving an $R^2 = 0.936$ as compared to $R^2 = 0.932$ with the original constant step size for the WC. This suggests that a step size of 0.25 Å already provides optimal resolution to the AP-RDF descriptor such that a finer step size does not provide much additional information to the model. Nonetheless, since the total number of descriptors does not change (339) when using the variable step size compared to the constant step size, we have opted to use the product AP-RDF with the variable step size for the remainder of the work.

The primary objective of this work is to determine the optimal chemical descriptor(s) to predict low-pressure CO_2 adsorption properties in MOFs and to examine whether they perform better than purely geometric descriptors. Geometric (Geo), bag-of-atoms (BOA), chemical motifs density (CMD), AP-RDF (RDF) descriptors, and combinations of these were investigated as descriptors for our ML models. From this point forward, the AP-RDF refers to the original product AP-RDF. Given in Table 2 is a summary of the results using various descriptors. It should be noted that only the best combinations

of descriptors are given in Table 2, while results with all permutations of the descriptors are given in the Supporting Information.

For the prediction of CO_2 working capacity, the model built using only the geometric descriptors performed significantly worse ($R^2 = 0.706$) than all other models built using a single type of chemical descriptor, which gave R^2 values of 0.838 or higher. The chemical motifs and the bag-of-atoms descriptors gave models with $R^2 = 0.838$ and 0.864, respectively. The single best chemical descriptor for predicting low-pressure CO_2 working capacity was the AP-RDF descriptor with an R^2 value of 0.936. Similar results were observed for the prediction of CO_2/N_2 selectivity. It is evident that chemical descriptors, even those that do not contain any geometric information, such as the bag-of-atoms descriptor, perform significantly better than the purely geometric descriptors when predicting low-pressure CO_2 adsorption properties. These results suggest that the chemistry of the pores is the dominant factor in determining the low-pressure CO_2 adsorption properties of MOFs and likely for other nanoporous materials. However, one significant disadvantage of the bag-of-atoms descriptor is that the descriptor is not invariant to rotations and replications of the MOF structure unlike the chemical motifs and AP-RDF descriptors.

Models were also built using combinations of the descriptor types. In all cases, the addition of the geometric descriptors to a set of chemical descriptors improved the results. The best performing model was built with a combination of the AP-RDF, geometric, and chemical motifs descriptors which gave a $R^2 = 0.958$ and 0.948 for WC and selectivity, respectively. It is notable that removal of the chemical motifs descriptor from the top performing model resulted in a very small decrease in performance with a model built using only the AP-RDF and geometric descriptors giving an R^2 of 0.953 and 0.943 for the WC and selectivity, respectively. One disadvantage of the chemical motif descriptor is that it is only effective in materials that contain the chemical motifs on which the model was trained. Thus, with a minimal decrease in performance, the model built with only the geometric and AP-RDF descriptors is more universally applicable.

Figure 3 shows heat maps of the ML-predicted values versus the actual (GCMC-simulated) values in the test data set for (a) CO_2 working capacity and (b) CO_2/N_2 selectivity. The colors of the heat map correspond to the density of data points in various regions of the graph, and the plots reveal that most of the MOFs in the test set are low-performing. In fact, approximately 50% of the materials in the test set have uptake

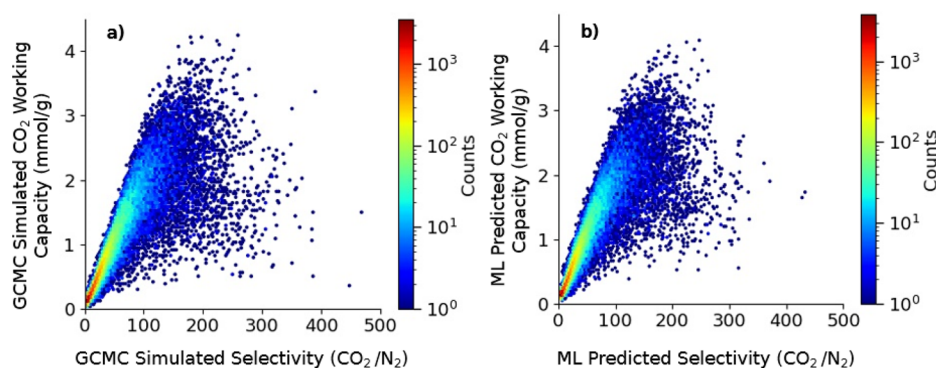


Figure 4. Heat maps of (a) GCMC-simulated and (b) ML-predicted CO₂ working capacity as a function of CO₂/N₂ selectivity for the test data set. The color of the heat map corresponds to the density of data points with highest density to lowest density being represented by the range of red to blue, respectively.

capacities of less than 0.40 mmol/g. Linear regression analysis reveals that the models have a tendency to slightly underestimate the WC and selectivity with best fit slopes of 0.96 and 0.93, respectively.

Figure 4 gives heat maps of CO₂ WC versus CO₂/N₂ selectivity for the (a) real GCMC data and (b) the ML-predicted values for the best ML models built from the geometric, AP-RDF, and chemical motif descriptors. It is generally accepted that MOFs with both high CO₂ WC and high selectivity make the best materials for postcombustion CCS,^{18–20} and comparing panels a and b of Figure 4 reveals that a very similar relationship of these two properties is found in the ML-predicted quantities compared to the actual data. This suggests that the ML model could be used to identify overall top-performing materials.

As previously mentioned, most of the MOFs in the data set have relatively low WCs and selectivities. Thus, it is conceivable that the high R^2 values of the models developed here are a result of their ability to accurately predict poorly performing MOFs. However, if the principal use of the models is to accelerate screening, then the models must be able to identify high-performing materials accurately. To examine this more closely, we selected the actual top 1000 MOFs in the test set (68 497 MOFs) as determined by the GCMC simulations. This subset is shown as the MOFs to the right of the solid red line in Figure 5. We then compared this subset to the top 1000 MOFs as predicted by our best ML model, which are the materials above the dashed red line in Figure 5. The intersection of these sets (the red shaded area) yields the top-performing MOFs that were recovered by the model. For the CO₂ working capacity, 773 of the actual top 1000 MOFs are recovered by the ML model's top 1000. The question now becomes how far down one must move the dashed line in Figure 5 to capture most or all the actual top-performing materials. This is given in Table 3, where the number of the actual top 1000 MOFs that are recovered in the top N -thousand MOFs as predicted by the ML models is reported for the various descriptor combinations. This procedure can be thought of as an analogue to recall in the assessment of classification models. For predicting the CO₂ working capacity, one can recover 994 of the top 1000 MOFs by sampling the top 5000 MOFs predicted by the best ML model. In other words, if one were to use the ML models as a prescreening tool, then to capture >99% of the true top 1000 MOFs, one would need to screen ~7% of the database with more compute intensive GCMC simulations. This would result in a greater

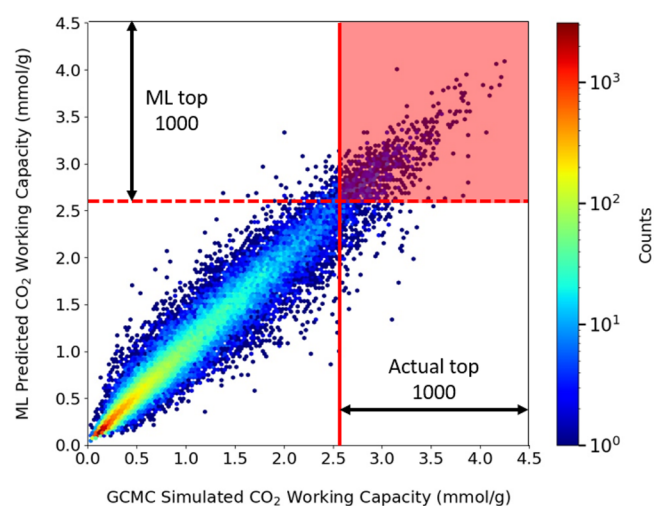


Figure 5. Heat map of the machine-learning-predicted CO₂ working capacity as a function of the GCMC simulated CO₂ working capacity for the best QSPR neural network model. The top 1000 MOFs predicted by the ML model are shown as those above the dashed line and the actual top 1000 MOFs as determined by GCMC simulation are shown as those to the right of the solid line. The intersection of these sets (the shaded area) represents the number of actual top-performing MOFs recovered by the ML model.

than 10× speed-up compared to screening all materials with GCMC simulations. Very similar results are found for the CO₂/N₂ selectivity models. We also note that when sampling the top 5000 MOFs from the ML-predicted top performers there is no significant difference between the screening performance of the best model built with the geometric, AP-RDF, and chemical motifs descriptors compared to the model built using only the geometric and AP-RDF descriptors. Furthermore, as a single descriptor, models built using the AP-RDF descriptor outperforms all other descriptors evaluated in this work. In fact, the AP-RDF-only model almost matches the screening performance of the best performing multitype descriptor models.

Although many ML regression models have been developed to predict high-pressure gas adsorption properties in nanoporous materials,^{30,31,38,51} there is a dearth of work on low-pressure (<1 bar) adsorption. In the earliest work, Fernandez and Woo developed SVM-based classifiers to categorize MOFs as being high- or low-performing for low-pressure CO₂ uptake. When applied in a prescreening scenario, those models could

Table 3. Number of Top-Performing MOFs Predicted by the Deep Learning Models for CO₂ Working Capacity and CO₂/N₂ Selectivity^a

target	descriptors	N = 1	N = 2	N = 3	N = 4	N = 5
CO ₂ working capacity	Geo	373	544	662	753	804
	BOA	553	739	839	888	914
	CMD	574	779	867	902	921
	RDF	722	910	957	976	981
	Geo+RDF	753	928	970	985	990
	Geo+CMD+RDF	773	943	975	986	994
	BOA	583	769	868	910	935
CO ₂ /N ₂ selectivity	Geo	423	645	783	885	936
	CMD	607	798	884	915	937
	RDF	749	928	959	982	991
	Geo+RDF	770	943	977	988	992
	Geo+CMD+RDF	776	933	967	978	987

^aOut of *N*-thousand MOFs that are in the actual top 1000 MOFs of the test set. The models were built using various combinations of geometric (Geo), bag-of-atoms (BOA), chemical motifs (CMD), and AP-RDF (RDF) descriptors.

recover 94% of the top-performing materials while flagging only 10% of the total database for more compute intensive screening.³⁶ The best model in the present work is able to capture 99% of the top-performing materials while flagging only 7% of the total database for more computer-intensive screening. Additionally, the models developed here were tested on a much more diverse library of materials. More specifically, the early work of Fernandez and others used a database constructed from 7 inorganic SBUs, 59 organic SBUs, and 19 functional groups, whereas the database used in the current is constructed from 23 inorganic SBUs, 175 organic SBUs, and 50 functional groups. Moreover, the original database spanned only 19 different network topologies, while the current database contains structures with over 1166 different network topologies.

More recently, Fanourgakis et al. introduced the Dprobes descriptor and used it to study low-pressure CO₂ adsorption on a subset of CoRE MOF database, containing 2932 structures.³⁵ At a pressure of 0.1 bar, they were able to predict CO₂ uptake with an *R*² value of 0.83 when the Dprobe descriptor was combined with geometric descriptors. Herein, the model built with the AP-RDF and geometric descriptors were able to obtain *R*² = 0.95 for the prediction of CO₂ working capacity at 0.15 bar, on a much larger and more diverse data set. It should be noted that the Dprobe descriptor is constructed by running a short atomistic simulation on the MOF using probe atoms consisting of varying van der Waals radii and electric dipoles. Although the atomistic simulation used to derive the Dprobes descriptor is short compared to a full GCMC simulation, the simulations require partial atomic charge parameters to be derived for each MOF. Unfortunately, deriving quality partial atomic charges can be the bottleneck in the GCMC workflow if they are derived from first-principles calculations as they were in that work. Furthermore, Fanourgakis et al. showed that accounting for electrostatic interactions between the probe atoms and the MOF are important in the case of low-pressure CO₂ adsorption.

CONCLUSIONS

In this work, deep learning neural networks were used with different descriptors to develop accurate quantitative structure–property relationship models to predict CO₂ adsorption properties at the low-pressure conditions relevant to postcombustion carbon capture (0.15 bar CO₂, 0.85 bar N₂). Several variations of the AP-RDF descriptor were evaluated, wherein the original product of atomic properties was replaced with the absolute difference in properties or the *sum* of the product of properties was replaced with the maximum. It was found that the original form of the AP-RDF descriptor performed the best and that combining the different variants of the AP-RDF provide minimal performance enhancement.

Models to predict CO₂ working capacity and CO₂/N₂ selectivity were then built using various combinations of geometric and chemical descriptors, including the original form of the AP-RDF descriptor. It was determined that geometric descriptors alone could not accurately predict low-pressure CO₂ adsorption, giving correlation *R*² values of only 0.71 and 0.75 for the uptake and selectivity, respectively. However, descriptors that accounted for chemical features of the MOFs provided significant improvement when used alone or in combination with geometric descriptors. This suggests that the chemistry of the pores is a dominating factor in determining the low-pressure adsorption of CO₂ and likely other gases. It is important to clarify that the geometric descriptors used in this work may not be the best geometric descriptors to use but rather are the most frequently used geometric descriptors. There are more information-dense geometric descriptors such as the recently developed topological descriptors which may perform better than the geometric descriptors used in this study when predicting low-pressure CO₂ adsorption properties of MOFs.^{52–54}

The most effective descriptor type was the AP-RDF, which provided exceptional model performance with *R*² = 0.94 in predicting the CO₂ working capacity. Importantly, the strong performance of the AP-RDF descriptor can partially be attributed to the fact that it captures geometric information on MOFs to a much greater extent than the other chemical descriptors investigated in this work. The best model was built using a combination of geometric, chemical motifs, and AP-RDF descriptors, with *R*² = 0.96 in predicting the CO₂ working capacity and *R*² = 0.95 for the CO₂/N₂ selectivity. These models have the highest *R*² scores for predicting low-pressure gas adsorption properties out of any models reported to date, with the caveat that a direct comparison to studies which use different databases cannot be reliably made. In addition to providing high-correlation *R*² values, the ML models were also shown to be able to identify the highest performing materials. More specifically, to capture 99% of the top 1000 MOFs from a data set of ~70 000 MOFs using the best model, one would only need to screen just 7% of the database with compute intensive GCMC simulations. When using the ML models as a prescreening tool during high-throughput screening to identify high-performing materials, this would result in a >10× speed up compared to using GCMC simulations alone. As such, these models have the potential to greatly accelerate high-throughput screening of materials and thereby compress the time lines of new material development.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jpcc.0c06334>.

More details regarding the development of the ML models and results for all combinations of descriptors. A list of the AP-RDF property parameters for each element. CSV file of all the descriptors and target values for all materials used to develop and test the ML models. Training, development, and test set CIFs (PDF)

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Notes

The authors declare no competing financial interest. The research data (and/or materials) supporting this publication can be accessed at <https://github.com/BurnerJ/descriptors-and-pytorch-models-for-low-pressure-co2-adsorption-in-mofs>.

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